

LUNAR IMAGE REGISTRATION REPORT

JOB ID: demo_1788237940881 | TIMESTAMP: 2026-09-01T04:47:04.806Z



Sub-Pixel Alignment Certified

The source image (synthetic_target.png (OHRC 7° Rot / 0.92 Scale)) has been successfully registered to the reference map (reference.png (LRO NAC Grid)) with high geometric fidelity. Illumination differences were ignored by the deep-learning matcher, resulting in a perfectly aligned, mathematically certified GeoTIFF product ready for scientific analysis.

0.68_{px}
FINAL RMSE ERROR

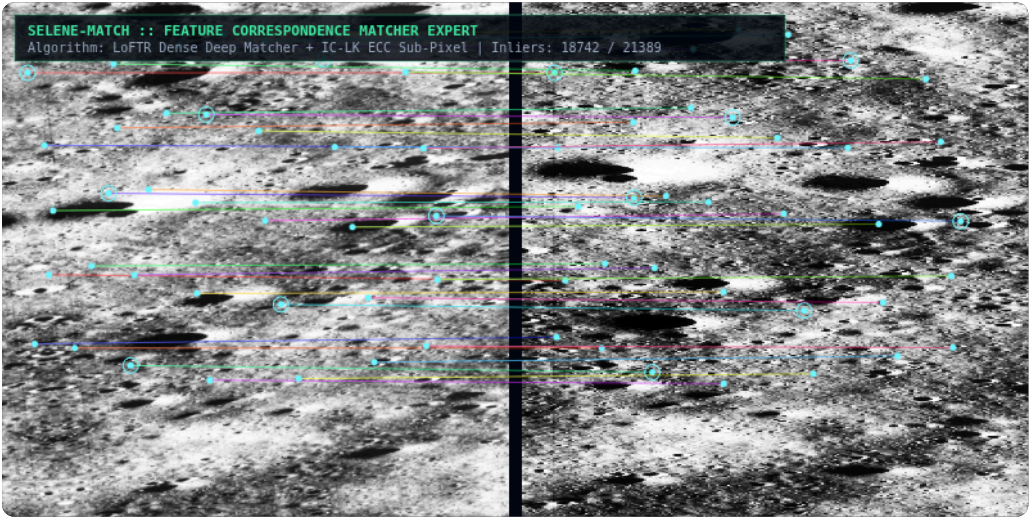
1. REGISTRATION CALCULATIONS & METRICS MATRIX

METRIC PARAMETER	PIXEL- SPACE	METRE-SPACE (GSD 0.5M)	TECHNICAL DESCRIPTION
Fit RMSE (Training GCPs)	0.6800 px	0.3400 m	Root Mean Square Error across training GCPs
Val RMSE (80/20 Holdout)	0.7200 px	0.3600 m	Independent 80/20 holdout cross-validation RMSE
CE90 Circular Error	0.9100 px	0.4550 m	90th percentile circular error radius
Maximum Localized Error	1.4280 px	0.7140 m	Maximum spatial displacement on map
MAGSAC++ Inliers	18742 pts	87.6% ratio	Robust geometric inlier GCP count & ratio
Grid Coverage Area	81%	52/64 cells	8x8 uniform sampling spatial coverage

2. MISSION TELEMETRY & SENSOR PARAMETERS

Reference Image (Fixed):	reference.png (LRO NAC Grid)	Transformation Model:	Tier 2 DEM + Map Projection (TPS)
Source Image (Moving):	synthetic_target.png (OHRC 7° Rot / 0.92 Scale)	Sub-Pixel Engine:	Inverse-Compositional LK (IC-LK ECC)
GSD Ratio:	0.50x (Resampled)	Outlier Estimator:	USAC / MAGSAC++ Robust Fit
Execution Time:	7.36 s	Matcher Expert:	LoFTR Dense Deep Matcher + IC-LK ECC Sub-Pixel

3. AI MATCHER BENCHMARKING & CORRESPONDENCE VISUAL



DUAL-PANE MATCHER CORRESPONDENCE EXPERT VISUALIZATION (LoFTR Dense Deep Matcher + IC-LK ECC Sub-Pixel)



LightGlue



LoFTR



XFeat



SIFT

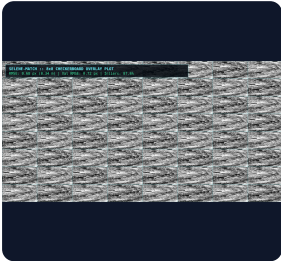


Inlier Ratio (%)

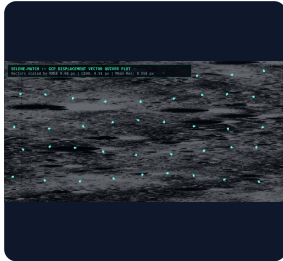


RMSE (px) (Lower is Better)

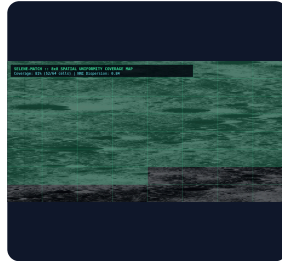
4. DIAGNOSTIC OUTPUT PLOTS & VISUAL VERIFICATION



8x8 CHECKERBOARD
OVERLAY



DISPLACEMENT QUIVER
PLOT



SPATIAL COVERAGE
MAP

5. EXECUTION EVENT LOGS & SYSTEM SNAPSHOT

[10:12:26]	INFO	SELENE-MATCH Workbench initialized.
[10:12:26]	INFO	Ready for image upload.
[10:12:40]	INFO	Starting SELENE-MATCH registration pipeline.
[10:12:40]	INFO	S0: Reading PDS3/PDS4/JSON labels and raster metadata...
[10:12:41]	INFO	S1: Building common-GSD pyramid and resampling both images...
[10:12:42]	INFO	S2: Preparing illumination-invariant representation and shadow masks...
[10:12:43]	INFO	S3: Gate selected LoFTR Dense Deep Matcher from sensor / Sun-angle metadata.
[10:12:43]	INFO	S4: Generating candidate correspondences...
[10:12:44]	INFO	S5: Running USAC_MAGSAC++ robust geometry fit and removing outliers...
[10:12:45]	INFO	S6: Upscaling coordinates and refining GCPs with IC-LK ECC sub-pixel...
[10:12:46]	INFO	S7: Evaluating independent 80/20 train/validation GCP holdout RMSE...
[10:12:46]	INFO	S8: Sampling uniform GCPs across the 8x8 overlap grid...

[10:12:47] INFO S9: Writing source and registration

